

FogNet Alive SIH26007

SIH26007: Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions
(Ministry of Steel / NMDC)

1. The One-Line Pitch

Every other fog-safety idea treats “the sensor network” and “the fleet” as two separate things — fixed nodes that sense, trucks that just listen. FogNet Alive removes that line. A truck that’s sitting idle can temporarily become a sensing node itself. A truck that’s driving through can check whether the last prediction it received was actually true, and quietly flag a sensor that’s gone bad. The network doesn’t just sit there predicting — it grows, shrinks, and corrects itself as the fleet moves through its normal shift, with almost no added hardware.

2. The Problem, Simply Stated

In an open-pit mine, trucks move along dirt roads carrying ore and people. When fog rolls in, a driver can lose sight of the road, other vehicles, or a bench edge (the steep drop-offs common in these mines) in seconds. Most existing systems only react once the driver is already inside the fog. FogNet’s job is to give that time back — by warning before, not after.

3. FogNet Alive — The Core Idea

This is the piece that makes FogNet different from a well-built version of the obvious idea. It has two parts. The first costs nothing extra to build. The second costs a small, specific amount, and we say so upfront rather than blur the two together.

3.1 Part One: Every truck verifies what it drives through (zero extra hardware)

Right now, a FogNode predicts fog is coming. Nobody ever checks if it was right. If a sensor drifts out of calibration, or gets covered in dust, it can keep broadcasting a wrong warning for weeks and nobody would know until a truck gets caught out by it.

When a truck drives through a FogNode’s zone, it already knows — as a free byproduct of the sensor fusion it’s already running — what visibility actually was, and whether the speed drop it needed matched what was predicted. FogNet Alive just broadcasts that one extra fact, tagged with which node it’s confirming and when.

The next truck approaching the same spot sees two things instead of one: the FogNode’s prediction, and the most recent truck’s lived confirmation of it.

- If they agree, the warning shows as **“confirmed 6 minutes ago by another vehicle”** — a live trust score, not just a model’s guess.
- If they disagree, the system doesn’t quietly average it away — it flags that specific FogNode

as possibly miscalibrated or blocked, and broadcasts that flag too, so a maintenance issue surfaces on its own instead of being found by a truck driving blind into a wrong prediction.

3.2 Part Two: An idle truck can temporarily become a node

A fixed FogNode is a category of hardware in every other design, including our own earlier version. Here, it's treated as a **role** — something a vehicle can hold temporarily, not something only one type of device can be.

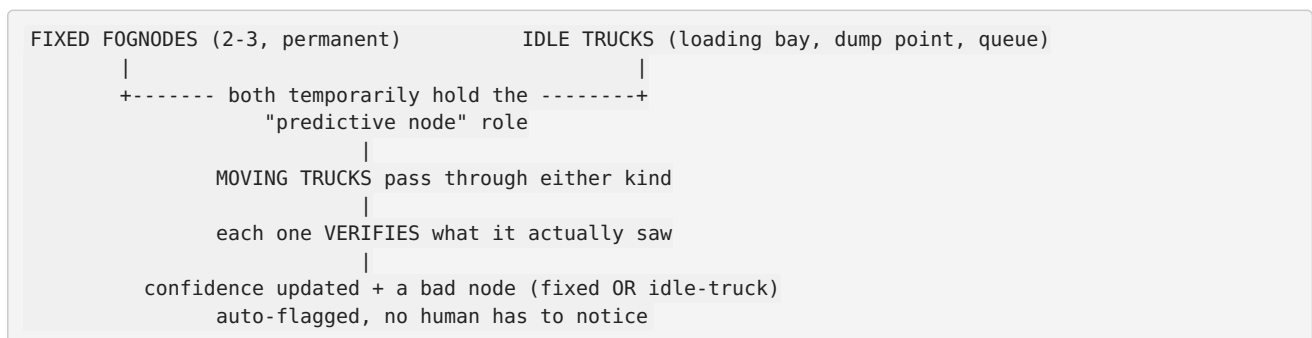
Trucks spend real chunks of every shift standing still: queued at a loading bay, waiting at a dump point, parked on a break — often at exactly the spots where trucks bunch up and risk is highest, and often not spots anyone budgeted a fixed FogNode for. The moment a truck's speed sensor shows it's been stationary for a couple of minutes, it quietly switches into node mode: watching conditions and broadcasting, the same job a fixed FogNode does. The moment it starts moving again, it switches back to being a normal vehicle. No installation, no site survey — your sensing coverage now follows wherever the fleet actually is, not just wherever you could afford to bolt something down.

The honest cost of the full version: a camera can tell an idle truck that fog has already arrived — useful, but reactive. A FogNode's real value is the 10-20 minute lead time, which comes from watching a humidity/temperature trend, not from a camera. To let an idle truck genuinely predict, not just confirm, it needs the same small humidity/temperature sensor a FogNode already carries — about Rs. 115-300 added per truck. Without that sensor, an idle truck can still verify and give reactive ground-truth (Part One, still useful, still free) — it just can't offer the early warning a fixed node gives. We say this distinction out loud before being asked, rather than letting a judge find the gap themselves.

3.3 Why this is hard for another team to copy quickly

This isn't a bolt-on feature — it only becomes possible once prediction, fusion, and V2V are all already solid, because it's built entirely out of data those three already produce. A team that hasn't already built that full stack can't add this in the time left at SIH; they'd have to build three subsystems first just to have the ingredients.

It also isn't a new idea sitting apart from the rest of FogNet — it's the same design instinct you'll see again in Alpha Mode (Section 13): a role, not a fixed identity. Alpha Mode already treats "escort" as something any capable truck can temporarily hold. FogNet Alive applies that same idea one level up, to the sensor network itself.



3.4 On patenting this — the honest version, not the hopeful one

"Vehicles share sensor data" by itself is not novel — connected-vehicle hazard sharing already

exists in the wider world. What's specific and worth protecting is the exact combination: an offline, GPS-free, battery-powered predictive mesh, where confirmation is a passive byproduct of safety fusion the vehicle runs anyway (not a separate reporting action), where a fixed sensing role can be temporarily assumed by any idle vehicle, and where mismatches self-flag a sensor's health — all with zero cloud or central server anywhere in the loop. That combination is narrow and specific enough to be worth a cheap Indian provisional filing before any public disclosure, including presenting at SIH, if you want a filing date protected. It is in the same tier of defensibility as Alpha Mode — a real, worthwhile claim to stake — not a guaranteed grant, and not a claim that nobody could ever independently arrive at the general idea.

4. Why FogNet Is Different — Updated

1. **It warns ahead of time, not in the moment**, through the FogNode network.
 2. **It checks its own predictions and grows its own coverage**, through FogNet Alive (Section 3) — a property almost no competing pitch at this problem statement will have, because it depends on already having built everything else well first.
 3. **Trucks look out for each other without needing to see each other**, sharing rough position, speed, and load state over radio.
 4. **Nothing has one single point of failure**. No control room, no GPS. Sensor failure leans on the network; network failure leans on the sensors; a bad fixed node gets caught by the fleet driving past it.
 5. **Every warning traces back to plain physics or a plainly stated rule** — never a black box.
 6. **It says what it doesn't do, before being asked**.
-

5. What We Deliberately Left Out (and Why)

No GPS. GPS regularly loses signal between the tall rock walls of an open pit. Trucks instead estimate how close another truck or node is using the strength of the radio signal itself (RSSI) — no extra hardware.

No central control room. A control room sounds reassuring but quietly becomes a single point of failure — if it goes down, every truck loses the same thing at once. Trucks and nodes talk directly to each other instead. The one real hazard a control room was solving — halting every vehicle at once for something like a blast — is covered instead by a small single-button Emergency Stop Beacon (Section 10).

No LiDAR. Fog and dust are clouds of tiny particles, and a laser scatters off every one of them before reaching the real obstacle — a well-documented general weakness of LiDAR in fog, not something specific to our setup. We use a camera plus ultrasonic instead — ultrasonic is sound-based and doesn't scatter the way light does.

6. System Overview

FogNodes (2-3 fixed) + Idle Trucks (temporary nodes, Section 3.2)

- > watch humidity, temperature, and light-scattering in the air
- > spot fog FORMING, 10-20 minutes before a truck arrives
- > broadcast a warning over LoRa radio (long-range, low-power, no internet needed)

Moving trucks

- > camera + ultrasonic, watching the road ahead
- > fuse both readings into one trusted picture
- > broadcast rough position, speed, and load state to nearby trucks
- > VERIFY what they actually saw against the last prediction (Section 3.1)
- > calculate a safe speed from stopping-distance physics
- > alert the driver: buzzer, LED, vibration
- > log everything for later review

Emergency Stop Beacon (one small unit, e.g. at a blast site)

- > one button -> highest-priority STOP message -> every truck in radio range

The FogNode/Alive network and the onboard reactive system run independently and only meet at the alert stage — if one half goes down, the other keeps working.

7. The Onboard Sensors — Layer by Layer

Sensor	What it does	Why this one
Camera	Detects obstacles and people (dehaze filter + YOLOv8-nano)	Cheapest source of real visual detail in normal conditions
Ultrasonic	Confirms very close objects (0.2–4 m) using sound	Doesn't scatter in fog the way light-based sensors do
LoRa radio	Sends/receives FogNode, Alive-node, and V2V broadcasts	Long range, low power, no cellular needed
Humidity/temp sensor (<i>FogNet Alive upgrade, optional</i>)	Lets an idle truck genuinely predict, not just confirm, fog trends	Same ~Rs. 115–300 part already used on fixed FogNodes

As fog thickens, the camera naturally becomes less reliable — the system automatically trusts the ultrasonic reading more. This is **confidence-weighted fusion**: each sensor's vote counts more or less depending on how trustworthy it currently is.

8. How Trucks Talk to Each Other (V2V) — and Who Goes First

Every truck radios its rough position, speed, and whether it's **loaded or unloaded** over the same LoRa link the FogNodes use.

A radio broadcast reaches every truck in range at the same instant — there's no way to make one truck hear it "first." What matters is what each truck's own system does with it.

Loaded trucks get priority — standard mine practice, not our invention. A loaded truck has a longer stopping distance and is harder to restart on a grade, so mine safety training already teaches that an empty truck should be the one to slow down or pull in. FogNet encodes that same rule: each broadcast carries a loaded/unloaded flag (a driver switch for the demo). If two trucks'

paths could cross, the loaded truck's display says "maintain speed, proceed," and the unloaded truck's says "yield, reduce speed."

Proximity is a rough three-level read (near/medium/far) from signal strength, never a precise number — in a rocky pit, reflections can distort it, so it's never fed into a fine-grained calculation, only coarse caution.

V2V is a helper, not the main safety system. The radio used here is built for range and low power, not guaranteed instant delivery, so onboard camera and ultrasonic sensing remain the primary way a truck avoids obstacles.

9. Communication Failure — What Happens If the Radio Link Drops

A link drops temporarily (a bend in the road, brief interference): every FogNode and truck broadcasts on a schedule, so a truck flags a source as "gone quiet" after a couple of missed intervals — not a full failure, just a note. Messages can be **relayed** — any truck or node that hears a warning repeats it once, so a truck doesn't need a direct line of sight to a distant FogNode if another truck sits between them.

The radio goes down completely: the truck falls back to exactly what it had before any networking was added — its own camera and ultrasonic sensors. It loses the early warning and other trucks' awareness, but never the ability to see what's directly ahead. The display shows "**NETWORK LOST — running on local sensors only**," with a distinct alert tone from an obstacle warning.

The one-line version for judges: *a sensor failing makes the truck lean on the network; the network failing makes it lean on the sensors — each weak point is covered by the other.*

10. Emergency Stop Beacon (Not a Control Room)

Blasting operations require every vehicle nearby to stop — today usually enforced with a siren a driver mid-task might miss. A control room used to be our answer to this, but it reintroduces the single point of failure we removed everywhere else.

Instead: a **single small radio unit with one button** (about the cost of a FogNode's radio). Pressing it broadcasts one **highest-priority STOP/CRAWL message** over the same mesh every truck already listens to — no dashboard, no operator watching a screen, nobody centrally in charge of the system's health. A decentralized trigger, not a centralized authority.

Honest scope: a Tier-3, only-if-time-allows build item, and an addition to existing blast-safety procedure (sirens, radio calls), not a replacement for it.

11. Fog Prediction — Now a Shared, Self-Checking Job

A FogNode — fixed or, temporarily, an idle truck (Section 3.2) — watches three things: how much light scatters in the air (the primary, direct signal), plus humidity and temperature as supporting evidence. Humidity rising alone doesn't trigger a warning; the scatter reading is what actually has to shift.

Adaptive sampling: by default, a node checks conditions slowly to save battery. The moment readings trend toward fog, it switches to fast sampling automatically — a plain rule, not a trained model, and the real reason this design can survive outside a hackathon table.

Verification (Section 3.1) runs on top of this everywhere, fixed node or idle truck alike: every prediction gets checked by the next truck that actually drives through, and a consistently wrong node — fixed or temporary — gets flagged without anyone having to notice by hand.

12. The Safe-Speed Calculation

FogNet continuously compares **distance needed to stop safely** (from current speed, reaction time, braking capability) against **distance actually available** (to an obstacle, another truck's broadcast position, or a fog-node warning zone). If the distance needed exceeds what's available, risk escalates and speed drops. This works correctly even for a stopped truck, or two trucks matching speed exactly — cases where a simpler "time to collision" number breaks down entirely.

If the calculated safe speed reaches zero and danger stays critical, our demo vehicle can cut its own motor power as a last resort. **We say clearly, before being asked: this is a small-scale proof-of-concept cutoff, not a certified braking system** — a real haul truck needs manufacturer integration and formal safety certification.

13. Sensor Failure and Alpha Mode — Role, Not Identity, Applied to Trucks

If a truck's own sensors fail, it drops to a crawl, switches to relying on nearby trucks' broadcasts only, shows "**SENSOR FAILURE — running on network data only**," and requires driver confirmation before resuming — the direct mirror of Section 9's network fallback.

Alpha Mode extends this to a bigger fleet. Two trucks are built with front-and-rear sensor coverage — "Alpha-capable" — but behave like any other truck until needed:

1. A truck's sensor health check reports failure.
2. The system checks whether reforming is safe right now (traffic, fog, blind curves). If not, the impaired truck uses ordinary single-neighbor fallback while the system keeps checking.
3. Once safe, a **formation advisory** — suggested position, speed, spacing — goes to the impaired truck and the two Alphas. A suggestion, not a command.
4. Drivers manually reposition when it's actually safe.
5. Once formed (confirmed via the same RSSI proximity read used everywhere else), the impaired truck gets guidance from both Alphas at once.
6. On recovery, the formation ends automatically and all three return to normal.

One shared display, not a special one. Every truck runs the same screen by default. An Alpha-capable truck shows one small extra panel only while actively escorting, which disappears the moment the escort ends — no permanent hierarchy in the UI, because there’s no permanent hierarchy in the fleet.

This is the same idea as FogNet Alive, applied to trucks instead of sensors: a role any capable unit can temporarily hold, not a fixed identity. That’s a real, repeated design language across the whole system, worth saying explicitly in your pitch — it’s not three unrelated clever tricks, it’s one idea applied twice.

14. Step-by-Step: How the Whole System Runs

1. **Sensing.** Camera and ultrasonic run continuously on every moving truck.
 2. **Advance warning.** FogNodes and any currently-idle trucks independently watch for fog forming and broadcast early.
 3. **Verification.** Any truck driving through checks the last prediction against what it actually saw, and broadcasts the result (Section 3.1).
 4. **Fusion.** Each truck combines its own sensor readings, trusting whichever is currently more reliable.
 5. **Awareness of other trucks.** Position, speed, and load state, shared over the same radio.
 6. **Decision.** The safe-speed engine compares stopping distance needed vs. available, factoring in obstacles, other trucks, and load-based priority.
 7. **Alert.** A clear speed recommendation plus buzzer/LED/vibration.
 8. **Fallback, if something fails.** Sensor failure leans on the network; network failure leans on the sensors.
 9. **Logging.** Every warning, verification, and fallback event is timestamped on each truck — a safety black box, not a central server.
-

15. Build Plan (36-Hour Hackathon Window)

Window	Target	Who
0-16h (must-have floor)	One rig: camera → dehaze → detection → ultrasonic fusion → safe-speed calc → buzzer/LED alert	All 3 AI/ML + 1 Electrical
16-26h (the differentiator — never skip)	2 FogNodes live, prediction running, message-relay logic (Section 9), and Part One of FogNet Alive — truck verification broadcasts (zero extra hardware)	ENTC + 1 Electrical, 1 AI/ML on prediction
26-34h (stretch, only if ahead)	Second rig for the V2V demo, load-priority logic, Alpha Mode, Emergency Stop Beacon, and — if time allows — Part Two of FogNet Alive: humidity/temp sensor on one rig so it can go into full idle-node predictive mode for the demo	Whoever finishes their track first

34–36h	Demo rehearsal, backup video, slide polish	Whole team
--------	--	------------

If Part Two doesn’t make it in time: Part One alone (verification, zero hardware) is still a real, demonstrable, judge-worthy feature on its own. Don’t let the more ambitious half put the guaranteed half at risk.

16. Bill of Materials (India, hackathon prototype)

Item	Notes	Approx. cost (Rs.)
Onboard rig #1 (camera, ultrasonic, compute, enclosure)	Laptop as compute, or Raspberry Pi 4	4,500–9,500
Onboard rig #2 (same, for the V2V demo)	Needed for two-truck scenes	4,500–9,500
LoRa module + antenna, one per rig	Same radio for FogNode, Alive, and V2V traffic	500 x 2 = 1,000
Relay + motor cutoff, one per rig	Last-resort auto-stop demo	150 x 2 = 300
2x FogNode (ESP32 + LoRa + humidity/temp + scatter pair)	~Rs. 1,200 per node	2,400
Emergency Stop Beacon (optional, Tier 3)	ESP32 + LoRa + button	~500
Subtotal (verification-only FogNet Alive, Part One)		≈ 12,700–23,200
Humidity/temp sensor, per rig (<i>FogNet Alive Part Two, optional</i>)	Enables full idle-node predictive mode	115–300 x up to 2 = 230–600
Total with full FogNet Alive		≈ 12,930–23,800

Part One (verification) adds no line item at all — pure software, reusing hardware and broadcasts already in the design. For comparison: commercial mine radar safety systems run Rs. 5–15 lakh per vehicle.

17. Demo Script (about 7 minutes)

- Clear conditions.** Both trucks moving normally, no warnings.
- Fog forms.** A fog machine activates near a FogNode. Both trucks’ displays show a warning **before either truck physically reaches the zone.**
- A truck goes idle and lights up as a node.** Park Truck 1 at a marked “loading bay” spot near the fog zone. Show its display switch to node mode and start broadcasting — call this out directly: *“this truck isn’t driving, but it’s still working for the rest of the fleet.”*
- Verification.** Send Truck 2 through the same zone shortly after. Show its display receive the FogNode’s prediction **and** Truck 1’s earlier confirmation, marked “confirmed Xm ago.”
- Two-truck collision avoidance — loaded vs. unloaded.** Send an unloaded Truck A and loaded Truck B toward each other through the fog on crossing paths. Neither camera can see the other. Show the loaded truck’s display read “maintain speed” and the unloaded truck’s

read “yield” — both slowing appropriately without ever seeing each other.

6. **Sensor failure.** Disconnect one truck’s camera mid-run. Show it drop to crawl and switch to network-data-only, while still keeping safe distance via radio.
7. **Communication failure.** Power off one FogNode or move a truck out of range. Show the affected truck flag “network lost” and fall back correctly to local-only operation.
8. **Last-resort auto-stop.** Place a static obstacle ahead of a truck until its safe speed reaches zero. Show the relay cut motor power, and state plainly: *“proof-of-concept cutoff on our demo vehicle — a real haul truck needs manufacturer brake integration and certification.”*
9. **Emergency Stop Beacon (if built).** Press the button; show both trucks halt immediately.
10. **Close.** Show the event log from the run, and close on the theme: *“every truck here did more than one job today — sensor, verifier, and for a few minutes, a node of its own. That’s not a feature we bolted on — it’s the same idea our Alpha Mode formation already uses, applied to the sensor network itself.”*

18. Honest Limitations — Said Before Judges Ask

- **Full idle-node prediction needs one added part per truck** (Section 3.2) — without it, an idle truck can still verify (Part One) but not predict ahead of time the way a fixed node can.
- **Verification needs enough trucks passing by to be useful.** With only two demo trucks, a “confirmed Xm ago” reading is a real result, but the value of this feature grows with fleet size — we’re honest that we’re demonstrating the mechanism, not fleet-scale statistics.
- **RSSI-based proximity is rough**, and rock walls in a real pit can distort it further — we only ever use it for coarse near/medium/far categories.
- **A fog-machine demo won’t calibrate exactly like real fog** — different particle size than natural fog, so the scatter sensor’s curve would need re-tuning on a real site.
- **FogNode (and idle-truck) battery life is a real question we haven’t fully quantified** — adaptive sampling is the right idea, but we don’t yet have a tested number for how many months a field battery would last.
- **Automatic stopping is a demo-scale safety backup, not a production braking system.**
- **This system supports the driver — it doesn’t replace one.** Every recommendation is advisory; a human always makes the final call.

19. Judge Questions — Prepared Answers

“What is FogNet Alive, in one sentence?” Every truck can verify what a sensor predicted, for free, and an idle truck can temporarily become a sensor itself — so the network gets more accurate and covers more ground the longer the fleet operates, without needing more fixed installations.

“Isn’t this just crowd-sourced hazard reporting — that already exists?” The general idea of vehicles sharing road conditions isn’t new. What’s specific here is the combination: it’s a byproduct of safety fusion the truck already runs, not a separate reporting action; it works fully offline with no GPS or cloud; and it lets a fixed sensing role be temporarily held by any idle vehicle, which then also self-flags a broken sensor without anyone checking by hand. That combination is what we’d consider defensible, not the broad idea of vehicles sharing data.

“Doesn’t a truck sitting idle to act as a node reduce productivity?” No — a truck only holds the node role while it’s already stopped for its own operational reasons (loading, queuing, a break). We’re not asking any truck to stop and wait; we’re using time that’s already being spent standing still.

“What if LoRa fails — how do FogNodes and trucks still talk?” A dropped link gets relayed by nearby trucks or handled by a timeout that flags it as “gone quiet.” A fully dead radio makes the truck fall back to its own camera and ultrasonic sensors — the same safety level as a stand-alone system, just without the early warning.

“With two trucks approaching a FogNode from opposite directions, who gets priority — loaded or unloaded?” The broadcast reaches both at the same instant — there’s no “first.” The loaded truck is advised to maintain speed and proceed; the unloaded truck is advised to yield, matching standard mine haul-road right-of-way practice.

“Do Alpha trucks get a special dashboard?” No — one shared display for every truck, with a temporary extra panel only while actively escorting.

“Why no control room?” It’s a single point of failure everywhere else in this design was built to avoid. The one real gap it filled — a supervisor stopping every truck at once — is covered by the Emergency Stop Beacon instead.

“Why not real mmWave radar, thermal cameras, or LiDAR?” Cost and build time for the first two; for LiDAR, fog and dust scatter its laser before it reaches the real obstacle — a well-documented general weakness, not specific to us.

“What about automatic braking on a real truck?” Explicitly out of scope — a decision-support and advisory system, not a certified braking system.

20. Team Roles

Role	Owns
AI/ML #1	Vision pipeline (dehazing + object detection)
AI/ML #2	Fog-onset prediction model, extended to run identically on idle-truck sensors (Section 3.2), plus adaptive sampling
AI/ML #3	Sensor fusion + safe-speed engine, V2V and load-priority input, sensor-failover logic, Alpha Mode state machine, verification/mismatch logic (Section 3.1)
Electrical #1	Onboard rig #1 hardware — camera/ultrasonic wiring, enclosure, relay/motor-cutoff
Electrical #2 (AI-PM / builder)	Onboard rig #2 hardware, humidity/temp sensor integration for FogNet Alive Part Two, driver display software (including node-mode and escort panels), Emergency Stop Beacon integration
ENTC	FogNode hardware, LoRa mesh protocol, message-relay and priority-flag logic, idle/moving role-switching logic, RSSI proximity estimation, range testing

21. Summary

FogNet Alive treats every truck as more than a vehicle: a sensor when it's moving, a verifier of what it just drove through, and — while it's idle anyway — temporarily a predictive node for everyone still on the road. Combined with a safe-speed engine grounded in real physics, a fleet that looks out for itself without a control room, and failure modes that quietly cover for each other, it's a system that gets more accurate and more capable the longer it runs — without needing anyone to install more hardware to make that happen.